



Organic Chemistry Reaction Mechanism

Applied chemistry (Pokhara University)



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SN1 reaction Continued from class

Reactivity of alkyl halides towards SN1 reaction:

The formation of carbocation is the rate determining step in SN1 reaction. So, the more stable of carbocation, the more easily it is formed and the more it facilitates the rate-determining step and speeds up the whole reaction. now the relative stability of the carbocation follows the order:

Tertiary > Secondary > Primary > Methyl carbocations

Therefore, the order of reactivities of alkyl halides towards the SN1 reaction follows the same order:

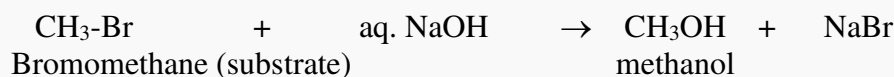
Tertiary > Secondary > Primary > Methyl halides

The relative stability of carbocations can be explained by the hyperconjugation effect.

Hyperconjugation is the partial interaction of sigma electron of C-H bond of an alkyl group directly attached to an atom of the unsaturated system or to an atom with unshared p-orbital.

SN2 reaction (bimolecular nucleophilic substitution reaction)

SN2 reaction stands for nucleophilic substitution reaction of 2nd order. In such reactions, rate determining step is depends on the concentration of only one species. Such reactions are generally shown by primary haloalkanes. For example, hydrolysis of methyl bromide with aq. NaOH (OH⁻)



Kinetics of SN2 reaction:

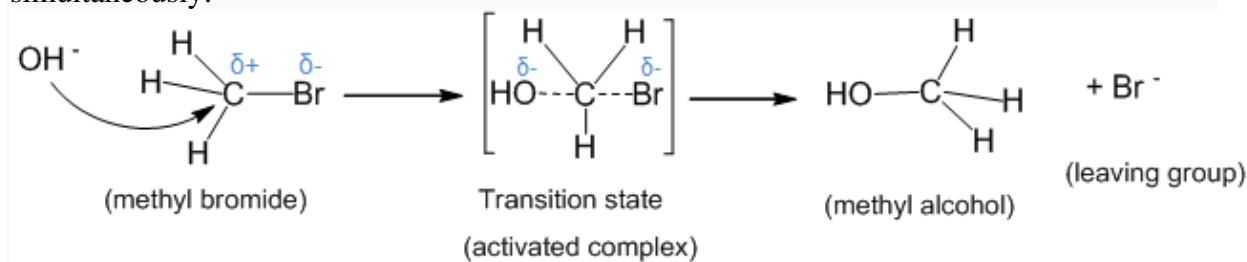
Since SN2 is reaction of 2nd order rate of the reaction depends upon the concentration of both reactants, called substrate (i.e. alkyl halide) and nucleophile and are present in rate determining step.

$$\text{Rate} = k [\text{CH}_3\text{-Br}] [\text{OH}^-]$$

Mechanism of SN2 reaction:

To explain the mechanism of SN2 reaction, let's take an example of hydrolysis of methyl bromide with aq. NaOH (OH⁻).

The mechanism of SN2 reaction involves a single step. Therefore, the breaking of carbon – halogen (C – Br) bond and formation of new carbon – nucleophile (C – OH) bond occurs simultaneously.



Stereochemistry of SN2 reaction

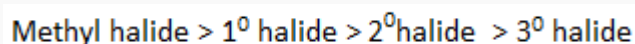
It is assumed that the nucleophile attacks the carbon atom attached to the halogen atom from the side opposite to the halogen (i.e. backside attack). As a result, a transition state (activated complex) is formed in which carbon atom is partially bonded to both nucleophile (-OH^-) and leaving group (-Br). In transition state, the central carbon is sp^2 hybridized and has triangular planar structure with bond angles of 120° .

Since activated complex is unstable, bromine leaves the molecule as a bromide ion and hydroxide ion forms a covalent bond with carbon to form alcohol with inverted configuration is formed as the product.

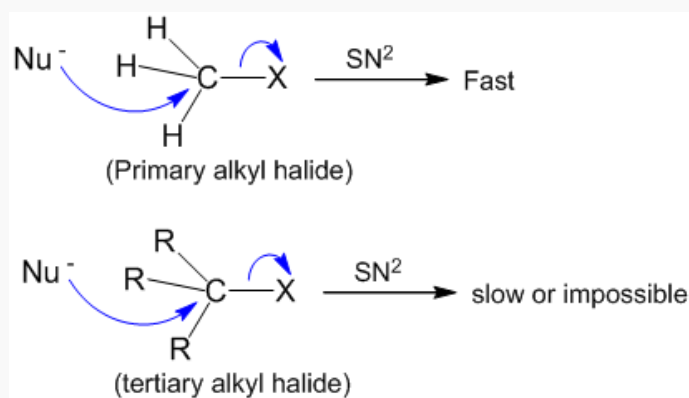
Reactivity of alkyl halides towards SN2 reaction:

The order of reactivities of alkyl halides towards the SN2 reaction is:

Reactivity of alkyl halides towards SN2 reaction



The reaction is faster when the alkyl group of the substance is methyl. When the hydrogen atoms of methyl group are replaced by bulkier alkyl groups, the increased crowding around central atom hinders the attack of nucleophile. This is called steric hindrance. Greater the steric hindrance, slower is the reaction.



The major differences between SN^1 and SN^2 reactions are as follows:

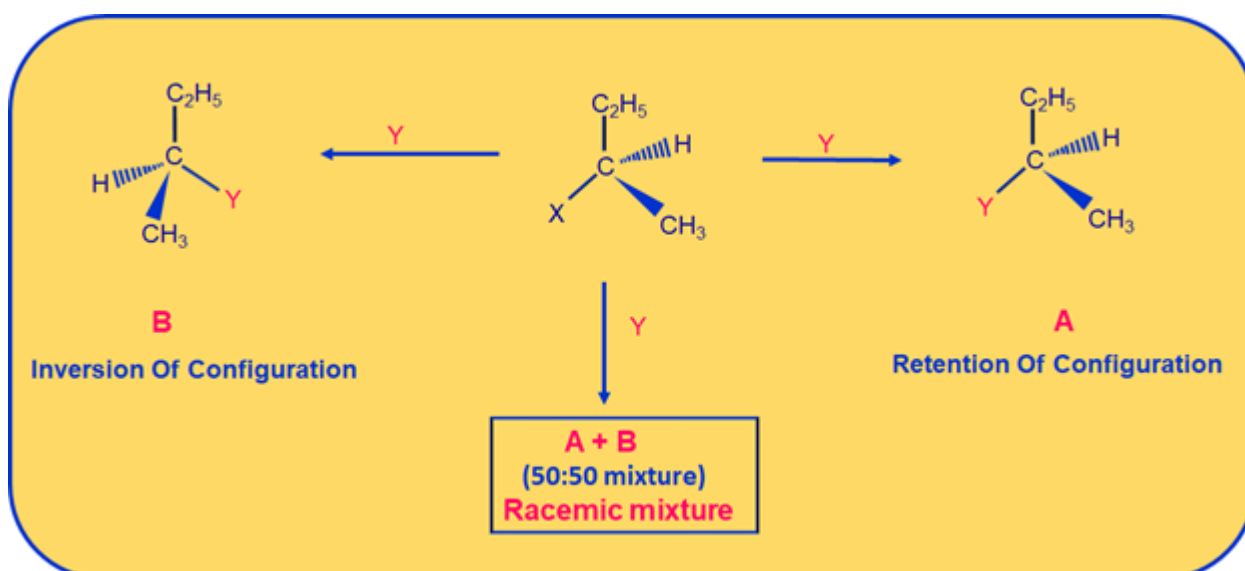
S.N	SN1 Reaction	SN2 reaction
1	It follows the 1 st order kinetics	It follows the 2 nd order kinetics
2	It is determined by electronic factors only: electron withdrawing and releasing group	It is determined by steric factors only: crowding of alkyl group around the carbon carrying halogen
3	The Nucleophile attack the carbocation from both sides although backside attack dominates	The Nucleophile attack the takes place from the back side only
4	Both Inversion and retention of configuring and hence Partial racemization of optically active halides takes place	Inversion of configuration takes place
5	It is favored by solvent of high polarity	It is favored by solvent of low polarity

6	Reengagements of products takes place	Reengagements of products does not takes place
7	It is favored by mild nucleophile	It is favored by strong nucleophile
8	The reactivity follows the order: $3^\circ > 2^\circ > 1^\circ > \text{methyl halide}$	The reactivity follows the order: methyl halide $> 1^\circ > 2^\circ > 3^\circ$

Notes: In **frontside attack**, the nucleophile attacks the stereocenter from the same side of the leaving group. This results in **Retention Of Configuration**. It means it is the preservation of integrity of the spatial arrangement of bonds to an asymmetric center during a chemical reaction.

In the **backside attack**, the nucleophile attacks the stereocenter from the opposite side of the leaving group. This results in formation of the product having **Inversion Of Configuration**.

Example:

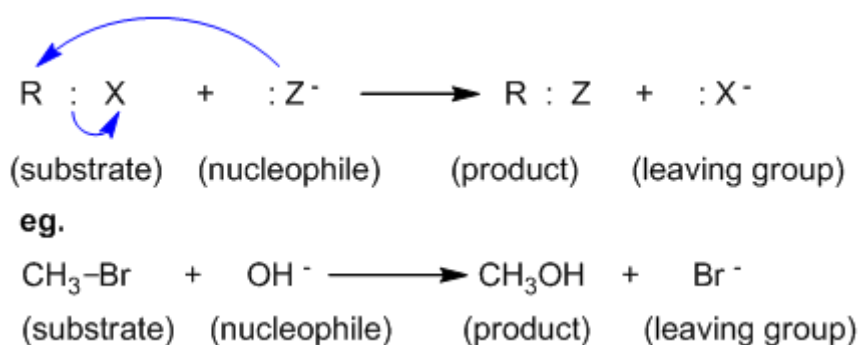


Electrophiles : (Electron lovers) . Electrophiles are electron deficient species which attack on electron rich centre during chemical reactions. These are either a positively charged species or the neutral species having electron deficient centre. Eg. Cl^+ , CH_3^+ , SO_3 , AlCl_3 , BF_3 , etc.

Nucleophile: A nucleophile is a reactant which donates an electron pair to form a covalent bond. It is electron rich species. Examples: RO^- , OH^- , RLi , $\text{RC}\equiv\text{C}^-$, and NH_2^- .

Substrate: it is a molecule or a compound that undergoes a chemical reaction, whether under the action of a catalyst, an enzyme or an inhibitor

Leaving group : an atom or group of atoms that detaches from the main or residual part of a substrate during a reaction or elementary step of a reaction. Common anionic leaving groups are halides such as Cl^- , Br^- , and I^- , water (H_2O), alcohols (HOR), and amines (R_3N) etc.



Factors that affect the S_N1 and S_N2 mechanisms:

- i. Nature of substrate
- ii. Nucleophilicity of the reagent
- iii. Solvent polarity

i. Nature of substrate:

a. **S_N2 mechanism:** The transition state (T.S.) of the S_N2 mechanism is pentacoordinate and thus crowded. As a result, S_N2 mechanism is favoured in primary halides and least favoured in tertiary halides. Reactivity order of alkyl halides towards S_N2 mechanism: $3^\circ < 2^\circ < 1^\circ < \text{CH}_3\text{X}$.

b. **S_N1 mechanism:** A planar carbocation intermediate is formed in the S_N1 reaction. It has no steric crowding as bulky alkyl groups can be easily accommodated in planar carbocation. Formation of planar carbocation intermediate results in a relief from steric crowding present in the tertiary halide substrate. Secondly, the carbocation intermediate is stabilized by +I effect of alkyl substituents and by the hyperconjugation effect of alkyl substituents containing α -hydrogens. As a result, S_N1 mechanism is most favoured in tertiary halides and least favoured in primary halides. Reactivity order of alkyl halides towards S_N1 mechanism: $3^\circ > 2^\circ > 1^\circ > \text{CH}_3\text{X}$.

ii. Nucleophilicity of the reagent:

a. **S_N2 mechanism:** A more powerful nucleophile attacks the substrate faster and favours the S_N2 mechanism.

b. **S_N1 mechanism:** The rate of the S_N1 mechanism is independent of the nature of nucleophile as nucleophile does not react in the slow step of S_N1. It waits till the carbocation intermediate is formed, and reacts fast with it.

iii. Solvent polarity:

a. **S_N2 mechanism:** Polar protic solvents usually decrease the rate of an S_N2 reaction. In the rate-determining step of the S_N2 mechanism, the substrate, as well as nucleophile, is involved.

A polar solvent stabilizes nucleophile (one of the reactants) by solvation. Thus, solvent deactivates the nucleophile by stabilizing it. Hence, aprotic solvents or solvents of low polarity will favour the S_N2 mechanism.

b. **S_N1 mechanism:** S_N1 mechanism proceeds via the formation of a carbocation intermediate. A good ionizing solvent, polar solvent, stabilizes the ions by solvation.

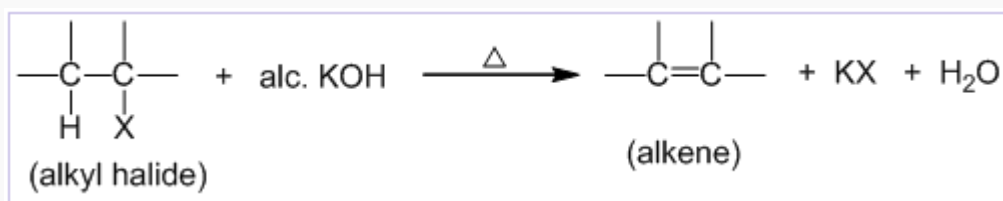
Solvation of carbocation is relatively poor and solvation of anion is particularly important. Anions are solvated by hydrogen-bonding solvents, that is, protic solvents.

Thus, the S_N1 reaction proceeds more rapidly in polar protic solvents than in aprotic solvents.

Elimination reaction

Any of a class of organic chemical reactions in which a pair of atoms or groups of atoms are removed from a molecule, usually through the action of acids, bases, or metals or heat are called elimination reaction.

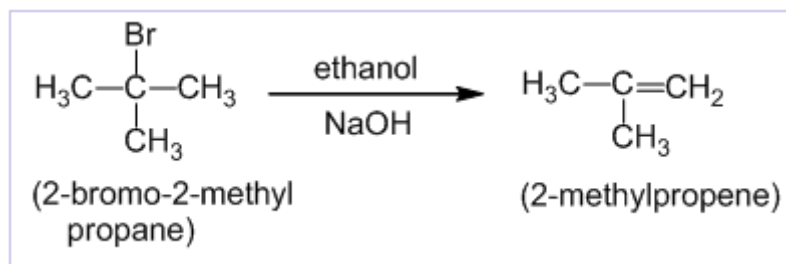
For example- Dehydrohalogenation reaction of alkyl halides.



When an alkyl halide is heated with an alcoholic solution of KOH or NaOH, an alkene is formed by the elimination of hydrogen atom from the carbon adjacent to the carbon-halogen linkage. So, this reaction is also called the α, β- elimination reaction.

E1 reaction

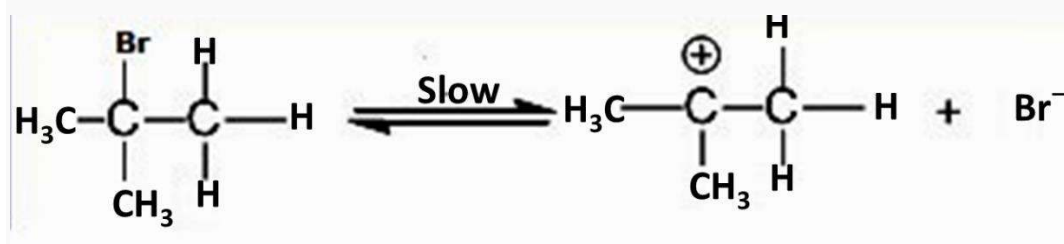
E1 reaction stands for the elimination reaction of first order (unimolecular reaction). This reaction is particularly common in secondary and tertiary alkyl halides in absence of a strong base. For example, when 2-bromo-2-methylpropane is treated with Alcoholic(ethanol) NaOH or KOH, 2-methyl propene is formed.



Mechanism of E₁ reaction:

The reaction is first order and thus occur in two steps.

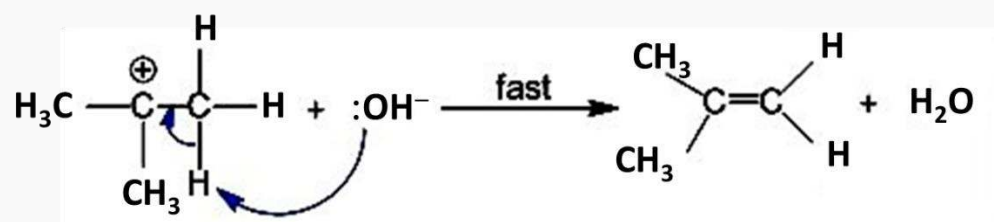
Step I : ionization of alkyl halide to form a carbocation (carbonium ion) and halide ion.



2-bromo-2-methylpropane
(tert-butyl bromide)

carbocation

Step II : elimination of a proton from the adjacent carbon(β-carbon) to yield the alkene.



2-methyl propene

The first step is slow step and is rate determining step, so first step is rate determining step of E₁ reaction.

In certain alkyl halides, the carbocation initially formed undergoes rearrangement to form more stable carbocation and thus highly branched alkene is formed as the major product.

Kinetics of E₁ reaction:

In the above E₁ reaction, the rate of alkene (2-methyl propene) formation depends upon the concentration of alkyl halide(tert-butyl bromide) only. It follows first order kinetics.

$$R = k [\text{tert-butyl bromide}]$$

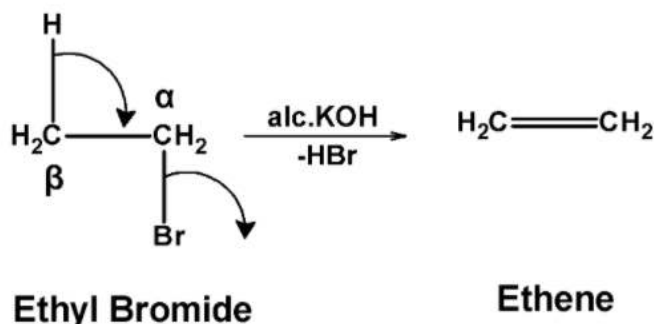
Reactivity of E₁ reaction:

Reactivity of alkyl halide to E₁ reaction is determined by the rate of formation of carbocation which in turn depends on stability of carbocation. Hence, order of reactivity of alkyl halides in E₁ reaction is:

3^0 haloalkane > 2^0 haloalkane > 1^0 haloalkane

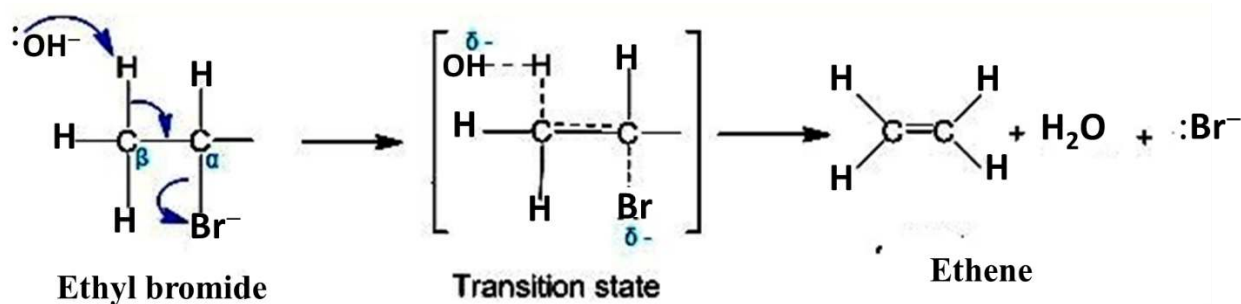
E2 reaction

E2 reaction stands for elimination reaction of the 2^{nd} order kinetics is also known as elimination bimolecular reaction. This reaction occurs when an alkyl halide is treated with a strong base such as hydroxide ion (OH^-). This reaction is particularly common in primary alkyl halides in presence of a strong base. For example, when ethyl bromide is treated with alcoholic NaOH or KOH, ethene is formed.



Mechanism of E2 reaction:

E2 mechanism is a one step process. As soon as base attacks the hydrogen atom of β -carbon, formation of carbon-carbon double bond and leaving of H-atom and -Br group starts to take place simultaneously as shown below in transition state. After the transition state, C-H bond and C-Br bond are completely broken and carbon-carbon double bond is formed.



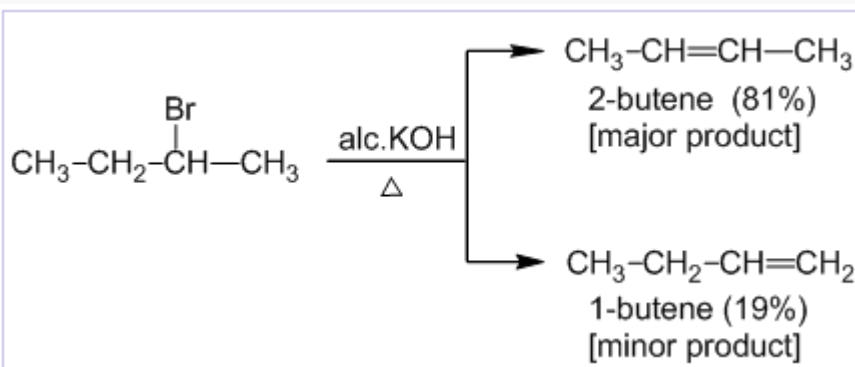
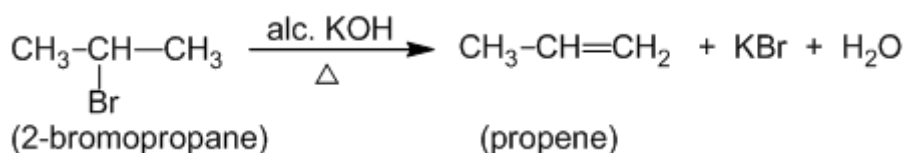
Kinetics of E2 reaction :

In E2 reaction, the rate of dehydrohalogenation (i.e. alkene formation) depends upon the concentration of both alkyl halide(ethyl bromide) and base(OH^-). It follows second order kinetics.

$$R = k[\text{ethyl bromide}] [\text{:OH}^-]$$

Orientation and reactivity of E₂ reaction:

In some cases, this reaction yields a single alkene but in other cases a mixture of alkenes are formed. For example, 2-bromopropane forms only propene whereas 2-bromobutane forms a mixture of 1-butene and 2-butene.



If there is a possibility of formation of mixture of alkenes, the E₂ reaction follows Saytzeff rule. This rule states that the major product is the alkene that has greater number of alkyl groups attached to the double bonded carbon atoms.

Therefore, the order of reactivity of alkyl halide in dehydrohalogenation is: 3° > 2° > 1°.

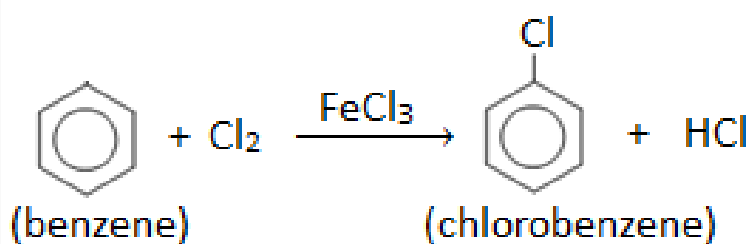
Electrophilic aromatic substitution reactions (benzene)

In electrophilic substitution reactions, an electrophile attacks the benzene and substitutes one of the hydrogen atoms of benzene ring to give substituted product.

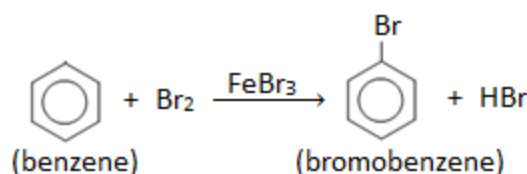


For example; **Halogenation of benzene**

Benzene reacts with halogen in presence of a Lewis acid such as FeCl₃ to give halobenzene. For example, chlorine reacts with benzene in presence of ferric chloride or AlCl₃ as catalyst to give chlorobenzene.



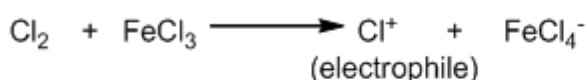
Similarly, benzene reacts with bromine in presence of ferric bromide as catalyst to give bromobenzene.



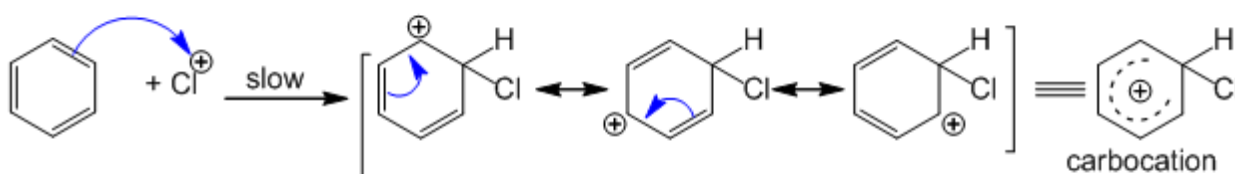
Mechanism of halogenation of benzene

This reaction involves the following steps:

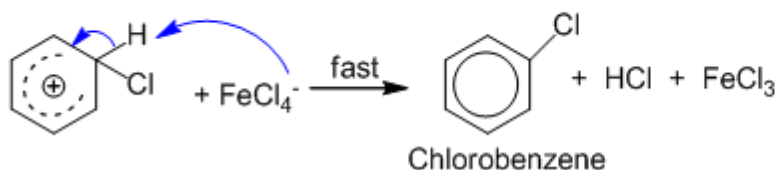
Step-I : Formation of electrophile.



Step-II: Reaction of electrophile with benzene to give intermediate cation.



Step-III : Reaction of cation with base to give substituted product i.e. chlorobenzene



Addition reaction

Addition reaction is a class of chemical reactions in which an atom or group of atoms is added to a molecule. Addition reactions are typical of unsaturated organic compounds—i.e., alkenes, alkynes, and aldehydes and ketones. During the reaction the double or triple bonds are fully or partially broken in order to accommodate additional atoms or groups of atoms in the molecule.

Types of addition reaction

- i) Nucleophilic addition reaction
- ii) **Electrophilic addition reaction**
- iii) **Free radical addition**

ii) Electrophilic addition reactions: Electrophilic addition reactions are brought about by electrophile and are typical reaction of alkene and alkyne. Electrophilic addition reaction follow the Markovnikov's rule.

For example; addition of HBr to propene.

Mechanism of electrophilic addition reaction

The addition of HBr to alkene takes place two steps

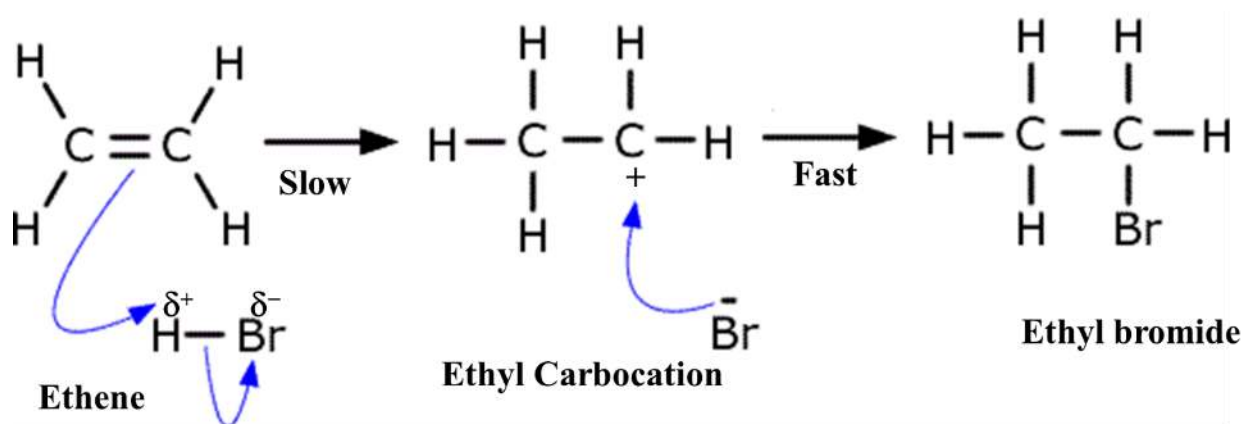
Step 1: formation of carbocation

The reaction starts with pi electrons attack on polarized electrophile and make a stable carbocation and the bromide ion.

This step is slow and hence is the rate determining step of the reaction.

Step 2. Nucleophilic attack by the bromide ion:

The carbocation formed immediately undergoes nucleophilic attack by the bromide ion present in the solution to form the addition product. This step is fast and hence does not affect the rate of the reaction.



Electrophilic addition of HBr to ethene

iii) free radical addition reaction:

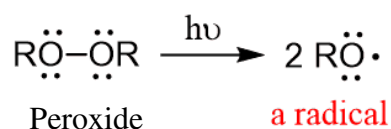
The addition reactions which involve free radicals are called free radical addition reaction. The reaction is governed by anti-markovnikov's rule.

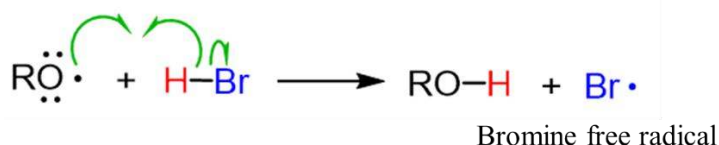
For example; addition of HBr to propene in presence of peroxide.

Mechanism of free radical mechanism

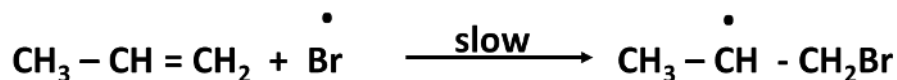
Let's take an example of addition of HBr to propene in presence of peroxide. The reaction takes place in 3 steps

i) Initiation: homolytic cleavage of the RO-OR(peroxide) bond. The resulting radical reacts with HBr by abstracting the H and a new radical Br[•] is formed:

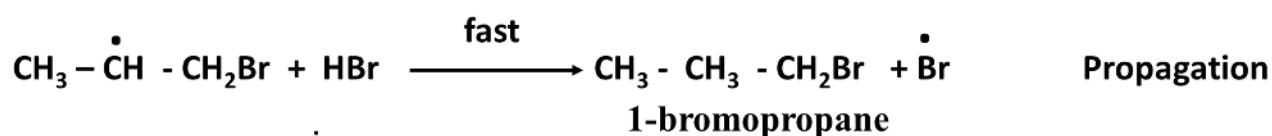




a regioselective addition of the Br to the alkene, the more substituted one forms as the major intermediate:



ii) propagation: Protonation of the highly substituted radical



iii) termination: combination of two bromine free radical to form Bromine molecule.



Reactivity of free radical addition of HBr to alkene is in the order: $3^\circ > 2^\circ > 1^\circ$

Note: regioselective: Regioselective in alkene addition reactions refers to which part the reagent adds to which part of the double bond.

Difference between Enantiomers and Diastereomers

Enantiomers	Diastereomers
1. Enantiomers are non-superimposable mirror images of each other.	1. Diastereomers are also non-superimposable but are not mirror images of each other.
2. All enantiomers have the same physical and chemical properties except optically active compounds	2. All diastereomers have different physical and chemical properties.

3. All Enantiomers pass optical activity. They have equal but opposite angles of rotation.	3. Not all Diastereomers pass the optical activity. They do not have an equal angle of rotation.
4. In the Enantiomers, the Shapes of molecules are similar.	4. Diastereomers have different molecular shapes.
5. Enantiomers are present in pairs.	5. There can be several molecules in Diastereomers.
6. They can not be separated by crystallization, chromatography, etc.	6. They can be separated by fractional distillation, chromatography, etc.
7. Example: Lactic acid	7. Example: Tartaric acid

Difference between SN1 and E1

SN1 reactions	E1 reactions
The SN1 reaction is defined as the unimolecular nucleophilic substitution reaction	The E1 reaction is defined as the unimolecular elimination reaction
In SN1 reaction there is a substitution of a weaker nucleophile	In E1 reaction there is an elimination of a functional group or atom.
SN1 reactions require a nucleophile in order to form the carbocation.	E1 reactions do not require a nucleophile to form the carbocation.
There is no formation of double bonds	There is a formation of double bonds
There is an involvement of one central carbon atom	There is an involvement of two adjacent carbon atoms
For example: reaction between tert-butyl bromide and aqueous NaOH or KOH to form tert-butyl alcohol.	For example: reaction between tert-butyl bromide and alcoholic NaOH or KOH to form 2-methyl propene.

Difference Between SN2 and E2 Reactions

SN2 reactions are nucleophilic substitution reactions that are bimolecular.	E2 reactions are elimination reactions in organic chemistry that are bimolecular reactions.
SN2 reactions are substitution reactions.	E2 reactions are elimination reactions.
SN2 reactions require a nucleophile.	E2 reaction does not require a nucleophile.
SN2 reactions do not require a base essentially.	E2 reactions require a strong base.

SN2 reactions prefer polar aprotic solvents.	E2 reactions prefer polar protic solvents.
The SN2 reaction rate is determined by the nucleophilic strength, solvent type, stability of the leaving group, etc.	The E2 reaction rate is determined by the strength of the base, solvent type, stability of the leaving group, etc.
For example, hydrolysis of methyl bromide with aq. NaOH (OH^-)	For example, when ethyl bromide is treated with alcoholic NaOH or KOH, ethene is formed.

Write down the Difference Between E1 and E2 Reactions

E1 vs E2 Reactions	
E1 reactions are a two-step elimination reactions	E2 reactions are a one-step elimination reactions
The E1 reaction occurs in either the complete absence of bases or in the presence of weak bases.	E2 reactions occur in the presence of strong bases.
The reaction mechanisms of E1 reactions are known as unimolecular eliminations.	The reaction mechanisms of E2 reactions are known as bimolecular eliminations.
E1 reactions are two-step reactions.	The E2 reaction mechanism is a single step elimination reaction.
E1 reactions form carbocations as intermediate compounds.	E2 reactions do not form any carbocation.
E1 reactions are known as unimolecular eliminations.	E2 reactions are known as bimolecular eliminations.
E1 reactions are common in tertiary alkyl halides and some secondary alkyl halides.	E2 reactions are common in primary alkyl halides and some secondary alkyl halides.
E1 follow the reactivity order of : Tertiary .> Sec. > Primary	E2 follow the reactivity order of : Primary > Secondary > tertiary.